

E7a near-tie sector agent

Algorithms, strategic ideas, evidence and failure profile

Arena identity agent 6590141, submission 41081503
Candidate 62,820 bytes; SHA-256 97bfe71e3f2f05e1...9fa9be79a8e48595
Stable parent 62,725 bytes; SHA-256 a8eb3b2bb646c59b...dbded047b884e55
Code delta +95 bytes; one complete function anchor, MoisanBot::focus_type
Mature checkpoint 25.34, rank 11/131, 160 games

What this agent is. The exact stable pre-seed/secure-orchard controller with one map-conditioned opening choice: prefer PLUM denial when PLUM is closer, or when its total access penalty relative to LEMON is at most eight path cells. The choice is frozen at initialization and affects a soft chop bonus. Training, pathfinding, assignment, collision handling, orchard protection and endgame conversion are inherited byte for byte.

1. Executive assessment

E7a is an unusually interpretable candidate. It does not add a wrapper, a new state machine or a learned model. It changes one strategic bit—which natural fruit species receives the denial bonus—only in a narrow opening geometry sector. This preserves the established parent on most maps and avoids the large loss of a global PLUM flip.

At the 160-game checkpoint it was the project's highest mature live scorer: 82 wins, three ties, 75 losses, score 25.34 and rank 11/131. The result is encouraging but not causal: the rule was selected on consumed development outcomes, its bootstrap interval crosses zero, and there is no same-window parent control.

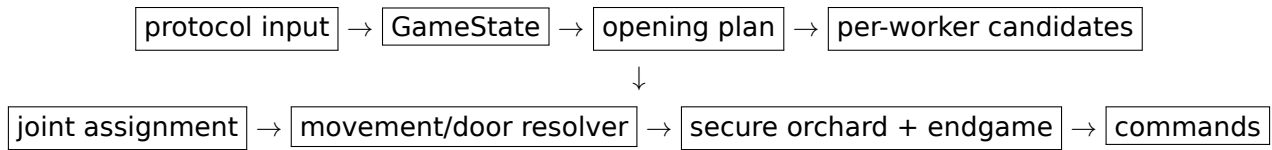
The bot is also not liveness-clean. A new exact-replay audit found period-2 MOVE episodes of at least six turns in 25/160 games. In game 897832286, a worker carrying two wood alternates between cells (6,3) and (6,4) from turns 160 through 286. The high ladder score therefore coexists with a material inherited execution defect.

2. Identity and evidence boundary

Artifact	Exact value
Candidate path	cgauto/submissions/ candidate-agent6553250-preseed-e7a-lemon-near-tie.min.rs
Candidate SHA-256	97bfe71e3f2f05e1b8fa3c697c5e5db3624ac9739e90954e9fa9be79a8e48595
Stable-parent path	cgauto/submissions/ candidate-agent6553250-preseed-orchard-coverage-slim.min.rs
Parent SHA-256	a8eb3b2bb646c59baf4c0a8b6bbdd9ca626e20ab2a27553dadbded047b884e55
Arena identity	agent 6590141; submission 41081503

The sacred development source was not modified. The live result is an owner-overridden exploratory deployment, not a prospectively qualified experiment. Local bridge evidence, consumed-panel pricing and live matchmaking are reported separately.

3. Overall controller architecture



The parser maintains terrain, units, plants, inventories and the turn counter. The inner controller produces scored actions for every worker. With two workers it exhaustively scores compatible candidate pairs. A movement layer reserves landing cells, applies orchard priorities and keeps shack doors usable. The outer secure-orchard wrapper may reserve one worker for an APPLE mother. Endgame logic converts fruit into wood when a full plant-chop-return cycle fits before turn 300.

The controller permanently caps itself at two workers. It is a refined two-worker denial and conversion policy, not a scalable economy.

4. The E7a near-tie sector rule

Let D_L be the sum of shortest-path distances from every walkable orthogonal door beside our shack to every natural LEMON tree. Let D_P be the corresponding PLUM sum. An unreachable tree contributes 10,000.

The stable parent chooses PLUM only if $D_P < D_L$; ties go to LEMON. E7a chooses PLUM when

$$D_P < D_L \quad \text{or} \quad 0 \leq D_P - D_L \leq 8.$$

It retains LEMON only when LEMON's aggregate path advantage exceeds eight. Therefore the patch changes parent behavior only in a LEMON-default near-tie sector. The result is cached as `type_to_cut` during opening; it is not recomputed each turn.

For a viable tree, base chop value is approximately

$$V_{cycle} = 1000 \frac{\text{expected wood}}{\text{travel} + \text{chop} + \text{return turns}}.$$

When the tree type equals `type_to_cut` and the rival has no more than two workers, the controller adds

$$V_{denial} = \frac{900}{1 + \text{Manhattan}(\text{tree}, \text{enemy shack})}.$$

This is a soft valuation shift. It does not command “chop PLUM”; a higher-value bank, mine, harvest or different tree may still win. The bonus is strongest close to the enemy shack, where denial has the greatest strategic intent.

5. Opening and workforce

The opening enumerates a second troll with movement, carry and chop in 1..3 and harvest fixed to zero. For each specification it estimates the ETA to collect the necessary PLUM, LEMON and, where present, IRON bill. It prefers carry at least two and chop at least one, uses a 15-turn horizon, accepts up to 15 turns of ETA slack for a stronger worker, and has a hard turn-35 downgrade deadline.

Before training, both trolls collect missing bill resources; carried material is banked instead of abandoned. At the deadline the controller chooses the strongest affordable fallback. After the second troll

is created, `can_train` always returns false for another worker. Recent top-player evidence makes this cap expensive: several stronger agents reach worker three in over 80% of games and worker four in roughly 40–95%.

6. Per-worker candidate generation

Banking. A carrier at a shack door receives a very high DROP candidate. Otherwise it receives a MOVE-to-bank candidate discounted by travel ETA.

Fruit. The controller predicts travel, tree ripening and required waiting. It issues HARVEST only on the tree. Persistent regenerative cargo is tracked as a commitment so a seed is not casually abandoned.

Iron. A worker routes to a walkable cell adjacent to ore and issues MINE when in position.

Wood. A chop candidate forecasts path length, growth and health, opponent chop damage, own chop duration, return path and final deposit. Unreachable, nonviable or too-late cycles are rejected. Throughput value is augmented by the E7a focus bonus.

Seeds. When resources are scarce after turn 100 and a worker is at a door, the controller may pick banked fruit for regeneration. In ordinary play, however, the policy remains predominantly chop-oriented.

7. Joint scheduling, collisions and doors

Candidates target a shack, bank, cell, tree or no exclusive object. Compatibility prevents two workers from claiming the same tree or cell and verifies that simultaneous PICKs do not exceed bank inventory. Because the bot caps itself at two workers, it evaluates the full Cartesian product and maximizes summed score rather than relying on unit order.

The movement resolver projects each landing cell at the unit's movement speed. Stationary units and earlier mover destinations are reserved. An orchard starter can receive priority, while the mother cell is forbidden to other units. A blocked landing triggers an orthogonal detour minimizing the remaining BFS distance; if no detour is legal, the worker waits.

A specialized door-clear layer handles carriers, blockers and swap patterns so that the own shack retains a usable door. These are strong local safety rules, but no global progress measure prevents two consecutive replans from reversing each other's target. The 160-game liveness result demonstrates the gap.

8. Secure APPLE orchard

The outer wrapper moves through Dormant, CarryingSeed, Active and Abandoned. It may establish one APPLE mother on an empty water-adjacent door cell near our shack. Activation requires at least two shack doors, distant natural trees, an enemy-safe mother, a trained second worker, an APPLE in the bank, alternate access for the other worker, and no sacrificed contested natural tree.

The starter picks the seed, plants the mother, deposits cargo, harvests ripe APPLES and otherwise waits. Other workers cannot occupy the mother or steal its seed. If ownership or safety fails, the wrapper abandons and returns to the inner controller.

This creates a protected renewable asset but can monopolize half the workforce. Across the new agent's 160 decoded games, only 10 of 1,704 attributed created crops were reaped by the agent. The live policy is therefore not a broad self-reproducing economy.

9. Endgame fruit-to-wood conversion

After turn 250—or when very few plants remain and the bot trails—fruit can be planted, chopped and returned before turn 300. A worker already on a tree receives very high chop priority. Ripe-fruit fallback is admitted only when travel, harvest, return and deposit all fit before the deadline.

This is a central behavior, not a corner case: the replay audit attributed 942 completed post-turn-250 planted-and-chopped conversions across the 160 games. It embodies a useful two-phase economic idea: renewable fruit is valuable early, while terminal wood is worth four points and dominates when no future harvest can amortize planting.

10. Why the sector idea was selected

Evidence	Result	Interpretation
Exploratory sector Association	13/60 roots OR 7.86	10 positive, 3 nonpositive; precision 76.92% Fisher $p = 0.00348$; discovery only
Exact bridge	16/16	inside sector = full flip; outside = parent; zero faults
Parent comparison	+4.008	root bootstrap 95% CI $[-1.588, +13.101]$
Score decomposition	+0.21 / − 3.80	own score / opponent score; gain mainly suppression
Global PLUM flip	−12.174	blanket policy is harmful
Hindsight oracle	+10.510	conditional opportunity exists but is not directly realizable

All six opponent-family means and all six leave-one-family-out means were positive, but the primary interval crosses zero. The scientific conclusion is narrower than the live rank: the rule is plausible, mechanically exact and sparse; its causal value remains unresolved.

11. Live behavior and the liveness defect

Games	160	Wins / ties / losses	82 / 3 / 75
Score	25.34	Rank	11 / 131
Mean margin	−29.3	Catastrophes (≤ -100)	35 (21.9%)
Negative mass	10,045	Runtime/identity faults	0

The stable parent's repeated mature median was 24.19 and best recorded mature run 24.77. E7a is +1.15 over the median and +0.57 over that best, but the comparison crosses eras and matchmaking populations. It is not an A/B estimate.

The new public-replay census decoded all 160 E7a games with zero fetch or state-decoder faults. Twenty-five games have a continuous period-2 MOVE run of at least six turns. The worst exact trace is game 897832286: worker 2 carries two wood and alternates $(6, 3) \leftrightarrow (6, 4)$ for 127 consecutive turns. Commands execute successfully, so this is planner liveness, not collision blocking or a runtime error.

12. Ideas worth retaining

1. **Tiny conditional patch.** Preserve a strong exact parent and alter one strategic choice rather than adding a broad wrapper.
2. **Geometry as policy context.** Convert a global denial decision into a map-sector decision.

3. **Aggregate access burden.** Sum route costs over the resource field instead of looking only at the nearest tree.
4. **Near-tie trade.** Accept a bounded own-efficiency penalty for a more useful denial target, but only where the costs are close.
5. **Frozen strategic intent.** Make the species choice at $t = 0$ to prevent high-level target churn.
6. **Behavioral bridge.** Prove exact parent behavior outside the sector and exact treatment behavior inside it.
7. **Mechanism decomposition.** Distinguish own production gain from opponent suppression; E7a's development gain is primarily the latter.

13. Limitations and next priorities

- The selection used consumed labels; no same-window parent control exists.
- The policy has a hard two-worker ceiling and no renewable macro-economy.
- Only PLUM versus LEMON is conditioned, and the decision never adapts after $t = 0$.
- Scores are hand tuned and deterministic; there is no learned opponent model.
- The orchard can reserve one of two workers indefinitely.
- The live tail is large: 35 catastrophes and 10,045 negative-margin mass.
- Legal movement is not sufficient for progress; long executed ABAB routes remain.

The best immediate engineering task is a baseline-preserving bank-commitment/liveness fix: once a wood carrier chooses to return, its destination and progress measure must remain stable until DROP or a proved obstruction. Validate against all 25 E7a counterexamples and the earlier ring counterexample before any value test. The larger research priority remains a scalable renewable economy with worker-three funding and concurrent production.

14. Reproducibility anchors

- Manifest: chatgpt_1/e7a-sector-candidate-manifest-2026-08-02.json
- Exact bridge: chatgpt_1/e7a-sector-candidate-bridge-2026-08-02.json
- Pricing: chatgpt_1/e7a-sector-candidate-pricing-2026-08-02.json
- Arena execution: data/analysis/live-agent-6553250/e7a-sector-owner-override-arena-execution-2026-08-02.md
- Top-15 inventory: data/analysis/live-agent-6553250/top15-public-battle-inventory-2026-08-02.json
- Full report source: docs/reports/2026-08-02-e7a-sector-agent-description.md